



NVIDIA RTX 3090 Spec Sheet

An in-depth look at the beefy specifications that lie beneath the glorious, black shell.
<http://dgit.in/nov20-47>



Time to boot-up Minecraft in 4K

Here's a list of Raytracing enabled games to test the might of the NVIDIA RTX 3080.
<http://dgit.in/nov20-48>

ZOTAC GeForce RTX 3090 Trinity Graphics Card

TITAN class gaming



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he RTX 3000 series had a rough launch. OEMs were allegedly not given enough time to test out the entire functionality of the GPUs. As a result some of the cards that reached the customers were affected by the now infamous capacitor fault. Some brands addressed the issue publicly without pointing fingers, while others denied any issues with their cards and certain brands quietly released hardware revisions to overcome the issue and one brand ended up with a card that had already fixed the issue thanks to a little inhouse R&D. If that wasn't enough, stocks of the new cards were barely present in the initial weeks and that will continue to be the case well into the next year. There have also been instances of scalping of these cards thanks to the insane performance increment. All that aside, some manufacturers did manage to get stocks of the card in India and ZOTAC is perhaps the only one to have enough cards. We managed to get our hands on the ZOTAC GeForce RTX 3090 Trinity and this is what we feel about it.



PERFORMANCE.....90
 VALUE.....75

tions, hashing out the latest cryptocurrency and gaming at 4K with all the graphics settings pushed to maximum. Boost clock goes up to 1695 MHz which is the same as the reference cards. So, you aren't getting

anything extra with the custom card aside from a different design. The reference card uses NVIDIA's new cooling mechanism while the ZOTAC Trinity uses a triple fan setup that we've seen on their flagship cards for the past several generations. It's a triple-slot graphics card and uses two 8-PIN PCIe power connectors. The rear is protected by a metal backplate while the shroud that houses the three cooling fans on the front is made of plastic.

A slight concern here is how the I/O plate is attached to the PCB. The card is attached to the I/O plate along one edge and there's a metal bracket that spans the length of the card providing a little extra reinforcement. For anyone who's studied trusses and how load is distributed across them, this particular method might cause slight uneasiness since this card is quite heavy and it does sag a little once installed.

Build

The ZOTAC GeForce RTX 3090 Trinity has 10496 shader units, 112 ROPs and is clocked at 1395 MHz. There's a massive 24 GB VRAM on the RTX 3090 so you could go about gaming, running big data computa-

Performance

In 3DMark, we prefer the Fire Strike Ultra benchmark since the Extreme and normal runs have started producing ridiculous scores with newer GPUs. The other benchmark within 3DMark we use is Time Spy and we

run both, the normal run and the extreme run. The RTX 3090 scores 12650 in 3DMark Fire Strike Ultra which puts it 55.8 per cent ahead of the RTX 2080 Ti and 18.6 per cent ahead of the RTX 3080. Time spy shows a slightly conservative figure putting the RTX 3090 about 2.4 per cent ahead of the RTX 3080.

Coming to gaming we ended up scoring 131 FPS in Borderlands 3, 207 FPS in Death Stranding, 319 FPS in DOOM Eternal, 169 FPS in Shadow of the Tomb Raider and 212 FPS in The Witcher III. To put this into perspective with the RTX 3080, we had scored 128 FPS in Borderlands 3, 205 FPS in Death Stranding, 305 FPS in DOOM Eternal, 164 FPS in Shadow of the Tomb Raider and 186 FPS in The Witcher III. Do note, these tests were redone after NVIDIA released new drivers to fix the capacitor issue. We see that the RTX 3090 doesn't seem to make a difference in a lot of video games and in some there's a moderate improvement. The card is a massive improvement over the RTX 2080 Ti but paying the extra premium for the RTX 3090 over the RTX 3080 makes little sense with these numbers.

Bottomline

The ZOTAC GeForce RTX 3090 Trinity Graphics Card is a great card if you have deep pockets. ZOTAC has largely managed to control the pricing of their graphics cards but one is better off getting an RTX 3080 or even an RTX 3070 since the latter equals the older RTX 2080 Ti. For roughly twice the cost of the RTX 3080, the RTX 3090 seems to be a card that gamers won't even bother with, but if you want a graphics card with an insanely massive VRAM cache, then that's a good reason to consider the RTX 3090.

—Mithun Mohandas

SPECIFICATIONS

CUDA CORES: 10496 | TENSOR CORES: 328 | RT CORES: 82
 | LITHOGRAPHY: 8nm Samsung | TDP: 350W | MEMORY:
 24GB GDDR6X | MEMORY BANDWIDTH: 936 GB/s.

CONTACT

ZOTAC | PHONE: NA | EMAIL: zkhan@zotac.com |
 WEBSITE: www.zotac.com